

1. Questions

Study the following information carefully and answer the questions.

The given table shows the cumulative number of cold tablets available, the cumulative number of fever tablets available, and the percentage of the number of pain tablets available out of the total number of tablets (cold + fever + pain) available in four different shops namely A, B, C and D.

Shop	The cumulative number of cold tablets available	The cumulative number of fever tablets available	The percentage of the number of pain tablets available out of the total number of tablets available
A	$X + 6$	$X + 84$	18.5%
B	$Y - 24$	$X + 309$	28.57%
C	$3X - 29$	Y	12.5%
D	$Y + 787$	$2Y - 382$	12%

Note:

- The ratio of the number of cold tablets to the number of fever tablets available in shop B is 7: 3 respectively.
- The first two columns of the table only show cumulative and the percentage of the number of pain tablets available in the respective shop out of the total number of tablets available in the respective shop.
- The number of cold tablets available in shop D is 155 more than that of shop C.

If the value of 'M' is $3Y - 6X$. Which of the following statements is true?

- The number of cold tablets available in shop C is ' $M - 5$ '.
 - The number of pain tablets available in shop A is '66.66% of M'.
 - The number of pain tablets available in shop B is ' $M - 22$ '.
- Both II and III
 - Both I and III
 - Both I and II
 - Only I
 - Only II

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2. Questions

In shop A, each strip of cold tablets contains 10 tablets, each strip of fever tablets contains 12 tablets, and each strip of pain tablets contains 6 tablets. Find the ratio of the number of cold, fever, and pain tablet strips, respectively

- a. 44: 43: 46
- b. 45: 44: 37
- c. 41: 39: 37
- d. 45: 43: 37
- e. 44: 41: 46

3. Questions

If the 60%, 80%, and 66% of the cold tablets, fever tablets, and pain tablets are sold in shop B. Find the total number of unsold tablets is what percentage of the total number of tablets available in shop B?

- a. 24%
- b. 38%
- c. 44%
- d. 28%
- e. 34%

4. Questions

If the sum of the number of cold tablets and fever tablets available in shop D is P and the sum of the number of cold tablets and pain tablets available in shop C is Q. Find the value of $(P - Q)$.

- a. $X + 246$
- b. $Y - 309$
- c. $Y - X + 135$
- d. All of them
- e. None of them

5. Questions

In shop E, the number of cold tablets is the average of those in shops A, B, and D. The number of fever tablets is 33.33% more than in shop C, and the ratio of the number of pain tablets E to D is 7:5. Find the difference in the total number of tablets between shops E and C.

- a. 328
- b. 210
- c. 368
- d. 486
- e. 1024

6. Questions

Study the following information carefully and answer the question.

A cycle race competition among 4 persons: Praveen, Nakul, Vamsi, and Suriya, and the race started at point A, which is 5 km away from point B, which is a finishing point. After 9 minutes of starting the race, Praveen travelled 600 m more than Nakul. After 12 minutes of starting the race, Praveen travelled 1.2 km more than Vamsi. The time taken by Nakul to finish the race is 100 sec earlier than that of Vamsi. The ratio of time taken by Vamsi to Suriya to travel 4.5 km is 5:3, respectively.

If Suriya reaches the finishing point B, at this time, find the difference between the distance travelled by Vamsi and that by Praveen.

- a. 1.2 km
- b. 1 km
- c. 1.5 km
- d. 2 km
- e. 0.8 km

7. Questions

If Nakul travelled after 9 minutes, he increased his cycle speed by 20%, then the total time taken by Nakul to reach the finishing point B in 'X' hour. Find the distance travelled by Vamsi in 'X' hour.

- a. 4.2 km
- b. 3.6 km
- c. 4.5 km
- d. 4.8 km
- e. 4 km

8. Questions

What is the ratio of the time taken by Nakul to that of Suriya to travel half of the racing distance?

- a. 2: 1
- b. 2: 5
- c. 1: 3
- d. 3: 5
- e. 3: 2

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9. Questions

If the speed of Adhi's cycle is the average of the speeds of Praveen and Nakul together. If Adhi travels N% of the total racing distance in 360 seconds, Find the value of 'N + 3.5N.'

- a. 192
- b. 188
- c. 172
- d. 198
- e. 182

10. Questions

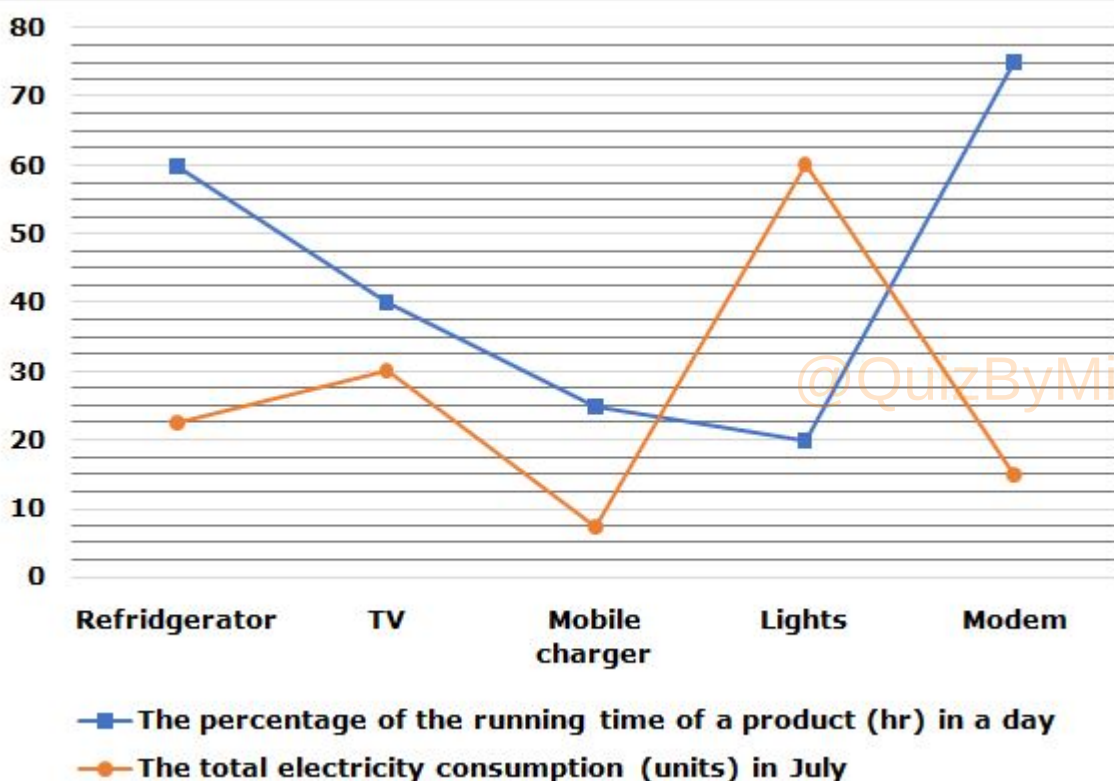
If Lara joined the race at his cycle speed, it is initially 20 km/hr; after 3 minutes, he increased his cycle speed by 20%, and after 4 minutes, again he increased his cycle speed by 50%. Who will finish the race first? (Remaining persons cycle speed is constant)

- a. Lara
- b. Praveen
- c. Vamsi
- d. Nakul
- e. Suriya

11. Questions

Study the following information carefully and answer the question.

The line graph below shows the percentage of the running time of a product in a day (hr) out of the total hours in a day and the total electricity consumption (units) in July in five different products in Nani home.



Note:

- i). The electricity consumption (units) per day = [the total electricity consumption (units) in July] ÷ [the running time of a product (hr) in a day]
- ii). The total electricity bill of July in Nani's home = the total electricity consumption (units) in July * per unit cost
- iii). Per unit cost is the first composite number.

In Nani's home, the total electricity consumption (units) in July is x% less than the previous month, and the total electricity bill of June and July together in Nani's home is Rs. 1215. Find the value of 'x'.

Note: per unit cost remains the same.

- a. 10
- b. 25
- c. 30
- d. 40
- e. 20

12. Questions

If a new AC is bought by Nani on July 1st and its running time in a day is 75% of that of the TV, the electricity consumption per day of AC is 5 units. Find the difference between the total electricity consumption (units) of AC and the total electricity consumption (units) of lights in July.

- a. 24 units
- b. 36 units
- c. 12 units
- d. 16 units
- e. 30 units

13. Questions

What is the ratio of the electricity consumption (units) per day on a refrigerator to the difference between the electricity consumption (units) per day on a TV and mobile charger?

- a. 4: 5
- b. 3: 4
- c. 5: 4
- d. 5: 6
- e. 1: 4

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14. Questions

If the total electricity consumption (units) of Kavi's home in July is 82% of that of Nani's home and the electricity consumption (units) of Mani's home in July is 33.33% more than that of Kavi's home. What is the average total electricity consumption for the three persons home together?

- a. 128.7
- b. 121.1
- c. 142.9
- d. 138.7
- e. 131.1

15. Questions

What is the total running time of all products (hr) in July? (All products running in the same units on all days)

- a. 1584
- b. 1636.8
- c. 1531.2
- d. 1320
- e. 1689.6

16. Questions

If the ratio of boys to girls in classes A and B is 4:3 and 3:2, respectively and $\frac{1}{10}$ of the boys and $\frac{1}{5}$ of the girls from the two classes come together to attend the poet competition. If 1 boy is selected from the competition, then the probability is $\frac{5}{12}$. Find the number of boys in class A.

Note: The total number of students in the two classes together is 85.

- a. 15
- b. 30
- c. 20
- d. 10
- e. 25

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17. Questions

The ratio of the efficiencies of A and C is 5:2. A and B together can complete the work in $7\frac{19}{23}$ days, while A, B, and C together can complete the same work in $5\frac{145}{151}$ days. If B and C start the work and continue for 5 days, after which C leaves and A joins, and A and B continue working for 't' days, then the remaining work is completed by B alone in 't + 7' days. Find the value of 't' -

t³.

- a. 27
- b. 9
- c. 18
- d. 3
- e. 54

18. Questions

The average weight of 18 boys in a class is $5x$ kg, and the average weight of 22 girls is $4x$ kg. If 10 students whose total weight is $34x$ kg leave the class, the average weight of the remaining students increases by 1.75 kg. The ratio of boys to girls among the students who left is 1:1 and the average weight of the boys who left is $(3x + 4)$ kg. Find the total weight of the girls who left the class.

- a. 75 kg
- b. 95 kg
- c. $(5x)$ kg
- d. Both (b) and (c)
- e. Both (a) and (c)

19. Questions

The ratio of the length, breadth, and height of the rectangular prism is 14:12:15, respectively. The total surface area of the rectangular prism is 4464 m^2 . If the radius of the cone is equal to the length of the rectangular prism and the height of the cone is equal to the breadth of the rectangular prism. Find the volume of the cone.

- a. 18512 m^3
- b. 18712 m^3
- c. 19512 m^3
- d. 19712 m^3
- e. None of these

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20. Questions

The total age of father, mother, and son in a family is 140 years. At the time of the son's birth, the father's age is 30 years, and the ratio of the present age of father and mother is 29:27, respectively. Find the present age of the grandfather, whose age is 3 times his grandson's.

- a. 84 years

- b. 58 years
- c. 54 years
- d. 72 years
- e. 78 years

Explanations:

1. Questions

The number of fever tablets available in shop B = $X + 309 - (X + 84) = 225$

The number of cold tablets available in shop B = $(225/3) * 7 = 525$

The percentage of the number of pain tablets available is $28.57\% = (2/7)$

So, The number of (cold + fever) tablets available in shop B to the number of pain tablets available in shop B = 5: 2

The number of pain tablets available in shop B = $(225 + 525)/5 * 2 = 300$

The total number of tablets available in shop B = $525 + 225 + 300 = 1050$

Then,

The number of cold tablets available in shop B = $Y - 24 - X - 6 = Y - X - 30$

So, $Y - X - 30 = 525$

$Y - X = 555 \rightarrow (1)$

Let,

The number of cold tablets available in shop C = $3X - 29 - Y + 24 = 3X - Y - 5$

The number of cold tablets available in shop D = $Y + 787 - 3X + 29 = Y - 3X + 816$

The number of cold tablets available in shop D is 155 more than that of shop C

$Y - 3X + 816 - (3X - Y - 5) = 155$

$2Y - 6X + 821 = 155$

$6X - 2Y = 666 \rightarrow (2)$

$(1) * 2$

$2Y - 2X = 1110$

Then, sum of equations (1) and (2)

$6X - 2X = 666 + 1110$

$4X = 1776$

$X = 444$

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And,

From (1)

$$Y - 444 = 555$$

$$Y = 999$$

In shop A,

$$\text{The number of cold tablets available} = X + 6 = 444 + 6 = 450$$

$$\text{The number of fever tablets available} = X + 84 = 444 + 84 = 528$$

$$\text{The number of pain tablets available} = [(450 + 528)/(100 - 18.5)] * 18.5 = [978/81.5] * 18.5 = 222$$

$$\text{The total number of tablets available} = 450 + 528 + 222 = 1200$$

In shop B,

$$\text{The number of cold tablets available} = Y - 24 - X - 6 = 999 - 444 - 30 = 525$$

$$\text{The number of fever tablets available} = X + 309 - X - 84 = 225$$

$$\text{The number of pain tablets available} = [(525 + 225)/(100 - 28.57)] * 28.57 = [750/(5/7)] * (2/7) = 300$$

$$\text{The total number of tablets available} = 525 + 225 + 300 = 1050$$

In shop C,

$$\text{The number of cold tablets available} = 3X - 29 - Y + 24 = 1332 - 999 - 5 = 328$$

$$\text{The number of fever tablets available} = Y - X - 309 = 999 - 444 - 309 = 246$$

$$\text{The number of pain tablets available} = [(328 + 246)/(100 - 12.5)] * 12.5 = [574/87.5] * 12.5 = 82$$

$$\text{The total number of tablets available} = 328 + 246 + 82 = 656$$

In shop D,

$$\text{The number of cold tablets available} = Y + 787 - 3X + 29 = 999 - 1332 + 816 = 483$$

$$\text{The number of fever tablets available} = 2Y - 382 - Y = 999 - 382 = 617$$

$$\text{The number of pain tablets available} = [(483 + 617)/(100 - 12)] * 12 = [1100/88] * 12 = 150$$

$$\text{The total number of tablets available} = 483 + 617 + 150 = 1250$$

Let table,

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Shop	The number of cold tablets available	The number of fever tablets available	The number of pain tablets available	The total number of tablets available
A	450	528	222	1200
B	525	225	300	1050
C	328	246	82	656
D	483	617	150	1250

Answer: C

According to the question,

$$M = 3 * 999 - 6 * 444$$

$$= 2997 - 2664$$

$$= 333$$

$$M = 333$$

I). The number of cold tablets available in shop C is 'M - 5'.

$$328 = 333 - 5$$

$$328 = 328$$

This is satisfied.

II). The number of pain tablets available in shop A is '66.66% of M'.

$$222 = 66.66\% \text{ of } 333$$

$$222 = 222$$

This is satisfied.

III). The number of pain tablets available in shop B is 'M - 22'.

$$300 = 333 - 22$$

$$300 \neq 311$$

This is not satisfied.

So, Both I and II are true.

2. Questions

The number of fever tablets available in shop B = $X + 309 - (X + 84) = 225$

The number of cold tablets available in shop B = $(225/3) * 7 = 525$

The percentage of the number of pain tablets available is $28.57\% = (2/7)$

So, The number of (cold + fever) tablets available in shop B to the number of pain tablets available in shop B = 5: 2

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The number of pain tablets available in shop B = $(225 + 525)/5 * 2 = 300$

The total number of tablets available in shop B = $525 + 225 + 300 = 1050$

Then,

The number of cold tablets available in shop B = $Y - 24 - X - 6 = Y - X - 30$

So, $Y - X - 30 = 525$

$Y - X = 555 \rightarrow (1)$

Let,

The number of cold tablets available in shop C = $3X - 29 - Y + 24 = 3X - Y - 5$

The number of cold tablets available in shop D = $Y + 787 - 3X + 29 = Y - 3X + 816$

The number of cold tablets available in shop D is 155 more than that of shop C

$Y - 3X + 816 - (3X - Y - 5) = 155$

$2Y - 6X + 821 = 155$

$6X - 2Y = 666 \rightarrow (2)$

$(1) * 2$

$2Y - 2X = 1110$

Then, sum of equations (1) and (2)

$6X - 2X = 666 + 1110$

$4X = 1776$

$X = 444$

And,

From (1)

$Y - 444 = 555$

$Y = 999$

In shop A,

The number of cold tablets available = $X + 6 = 444 + 6 = 450$

The number of fever tablets available = $X + 84 = 444 + 84 = 528$

The number of pain tablets available = $[(450 + 528)/(100 - 18.5)] * 18.5 = [978/81.5] * 18.5 = 222$

The total number of tablets available = $450 + 528 + 222 = 1200$

In shop B,

The number of cold tablets available = $Y - 24 - X - 6 = 999 - 444 - 30 = 525$

The number of fever tablets available = $X + 309 - X - 84 = 225$

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The number of pain tablets available = $[(525 + 225)/(100 - 28.57)] * 28.57 = [750/(5/7)] * (2/7) = 300$

The total number of tablets available = $525 + 225 + 300 = 1050$

In shop C,

The number of cold tablets available = $3X - 29 - Y + 24 = 1332 - 999 - 5 = 328$

The number of fever tablets available = $Y - X - 309 = 999 - 444 - 309 = 246$

The number of pain tablets available = $[(328 + 246)/(100 - 12.5)] * 12.5 = [574/87.5] * 12.5 = 82$

The total number of tablets available = $328 + 246 + 82 = 656$

In shop D,

The number of cold tablets available = $Y + 787 - 3X + 29 = 999 - 1332 + 816 = 483$

The number of fever tablets available = $2Y - 382 - Y = 999 - 382 = 617$

The number of pain tablets available = $[(483 + 617)/(100 - 12)] * 12 = [1100/88] * 12 = 150$

The total number of tablets available = $483 + 617 + 150 = 1250$

Let table,

Shop	The number of cold tablets available	The number of fever tablets available	The number of pain tablets available	The total number of tablets available
A	450	528	222	1200
B	525	225	300	1050
C	328	246	82	656
D	483	617	150	1250

Answer: B

The number of cold tablet strips = $450/10 = 45$

The number of fever tablet strips = $528/12 = 44$

The number of pain tablet strips = $222/6 = 37$

Thus, ratio = 45: 44: 37

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3. Questions

The number of fever tablets available in shop B = $X + 309 - (X + 84) = 225$

The number of cold tablets available in shop B = $(225/3) * 7 = 525$

The percentage of the number of pain tablets available is $28.57\% = (2/7)$

So, The number of (cold + fever) tablets available in shop B to the number of pain tablets available in shop B = 5: 2

The number of pain tablets available in shop B = $(225 + 525)/5 * 2 = 300$

The total number of tablets available in shop B = $525 + 225 + 300 = 1050$

Then,

The number of cold tablets available in shop B = $Y - 24 - X - 6 = Y - X - 30$

So, $Y - X - 30 = 525$

$Y - X = 555 \rightarrow (1)$

Let,

The number of cold tablets available in shop C = $3X - 29 - Y + 24 = 3X - Y - 5$

The number of cold tablets available in shop D = $Y + 787 - 3X + 29 = Y - 3X + 816$

The number of cold tablets available in shop D is 155 more than that of shop C

$Y - 3X + 816 - (3X - Y - 5) = 155$

$2Y - 6X + 821 = 155$

$6X - 2Y = 666 \rightarrow (2)$

$(1) * 2$

$2Y - 2X = 1110$

Then, sum of equations (1) and (2)

$6X - 2X = 666 + 1110$

$4X = 1776$

$X = 444$

And,

From (1)

$Y - 444 = 555$

$Y = 999$

In shop A,

The number of cold tablets available = $X + 6 = 444 + 6 = 450$

The number of fever tablets available = $X + 84 = 444 + 84 = 528$

The number of pain tablets available = $[(450 + 528)/(100 - 18.5)] * 18.5 = [978/81.5] * 18.5 = 222$

The total number of tablets available = $450 + 528 + 222 = 1200$

In shop B,

The number of cold tablets available = $Y - 24 - X - 6 = 999 - 444 - 30 = 525$

The number of fever tablets available = $X + 309 - X - 84 = 225$

The number of pain tablets available = $[(525 + 225)/(100 - 28.57)] * 28.57 = [750/(5/7)] * (2/7) = 300$

The total number of tablets available = $525 + 225 + 300 = 1050$

In shop C,

The number of cold tablets available = $3X - 29 - Y + 24 = 1332 - 999 - 5 = 328$

The number of fever tablets available = $Y - X - 309 = 999 - 444 - 309 = 246$

The number of pain tablets available = $[(328 + 246)/(100 - 12.5)] * 12.5 = [574/87.5] * 12.5 = 82$

The total number of tablets available = $328 + 246 + 82 = 656$

In shop D,

The number of cold tablets available = $Y + 787 - 3X + 29 = 999 - 1332 + 816 = 483$

The number of fever tablets available = $2Y - 382 - Y = 999 - 382 = 617$

The number of pain tablets available = $[(483 + 617)/(100 - 12)] * 12 = [1100/88] * 12 = 150$

The total number of tablets available = $483 + 617 + 150 = 1250$

Let table,

Shop	The number of cold tablets available	The number of fever tablets available	The number of pain tablets available	The total number of tablets available
A	450	528	222	1200
B	525	225	300	1050
C	328	246	82	656
D	483	617	150	1250

Answer: E

The number of cold tablets unsold in shop B = 40% of $525 = 210$

The number of fever tablets unsold in shop B = 20% of $225 = 45$

The number of pain tablets unsold in shop B = 34% of $300 = 102$

The total unsold tablets in shop B = $210 + 45 + 102 = 357$

Thus, percentage = $(357/1050) * 100 = 34\%$

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4. Questions

The number of fever tablets available in shop B = $X + 309 - (X + 84) = 225$

The number of cold tablets available in shop B = $(225/3) * 7 = 525$

The percentage of the number of pain tablets available is $28.57\% = (2/7)$

So, The number of (cold + fever) tablets available in shop B to the number of pain tablets available in shop B = $5:2$

The number of pain tablets available in shop B = $(225 + 525)/5 * 2 = 300$

The total number of tablets available in shop B = $525 + 225 + 300 = 1050$

Then,

The number of cold tablets available in shop B = $Y - 24 - X - 6 = Y - X - 30$

So, $Y - X - 30 = 525$

$Y - X = 555 \rightarrow (1)$

Let,

The number of cold tablets available in shop C = $3X - 29 - Y + 24 = 3X - Y - 5$

The number of cold tablets available in shop D = $Y + 787 - 3X + 29 = Y - 3X + 816$

The number of cold tablets available in shop D is 155 more than that of shop C

$Y - 3X + 816 - (3X - Y - 5) = 155$

$2Y - 6X + 821 = 155$

$6X - 2Y = 666 \rightarrow (2)$

$(1) * 2$

$2Y - 2X = 1110$

Then, sum of equations (1) and (2)

$6X - 2X = 666 + 1110$

$4X = 1776$

$X = 444$

And,

From (1)

$Y - 444 = 555$

$Y = 999$

In shop A,

The number of cold tablets available = $X + 6 = 444 + 6 = 450$

The number of fever tablets available = $X + 84 = 444 + 84 = 528$

The number of pain tablets available = $[(450 + 528)/(100 - 18.5)] * 18.5 = [978/81.5] * 18.5 = 222$

The total number of tablets available = $450 + 528 + 222 = 1200$

In shop B,

The number of cold tablets available = $Y - 24 - X - 6 = 999 - 444 - 30 = 525$

The number of fever tablets available = $X + 309 - X - 84 = 225$

The number of pain tablets available = $[(525 + 225)/(100 - 28.57)] * 28.57 = [750/(5/7)] * (2/7) = 300$

The total number of tablets available = $525 + 225 + 300 = 1050$

In shop C,

The number of cold tablets available = $3X - 29 - Y + 24 = 1332 - 999 - 5 = 328$

The number of fever tablets available = $Y - X - 309 = 999 - 444 - 309 = 246$

The number of pain tablets available = $[(328 + 246)/(100 - 12.5)] * 12.5 = [574/87.5] * 12.5 = 82$

The total number of tablets available = $328 + 246 + 82 = 656$

In shop D,

The number of cold tablets available = $Y + 787 - 3X + 29 = 999 - 1332 + 816 = 483$

The number of fever tablets available = $2Y - 382 - Y = 999 - 382 = 617$

The number of pain tablets available = $[(483 + 617)/(100 - 12)] * 12 = [1100/88] * 12 = 150$

The total number of tablets available = $483 + 617 + 150 = 1250$

Let table,

Shop	The number of cold tablets available	The number of fever tablets available	The number of pain tablets available	The total number of tablets available
A	450	528	222	1200
B	525	225	300	1050
C	328	246	82	656
D	483	617	150	1250

Answer: D

$P = 483 + 617 = 1100$

$Q = 328 + 82 = 410$

Thus, $P - Q = 1100 - 410 = 690$

a) $X + 246$

Let,

$444 + 246 = 690$

This is satisfied.

b) $Y - 309$

Let,

$999 - 309 = 690$

This is satisfied.

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c) $Y - X + 135$

Let,

$$999 - 444 + 135 = 690$$

This is satisfied.

5. Questions

The number of fever tablets available in shop B = $X + 309 - (X + 84) = 225$

The number of cold tablets available in shop B = $(225/3) * 7 = 525$

The percentage of the number of pain tablets available is $28.57\% = (2/7)$

So, The number of (cold + fever) tablets available in shop B to the number of pain tablets available in shop B = 5: 2

The number of pain tablets available in shop B = $(225 + 525)/5 * 2 = 300$

The total number of tablets available in shop B = $525 + 225 + 300 = 1050$

Then,

The number of cold tablets available in shop B = $Y - 24 - X - 6 = Y - X - 30$

So, $Y - X - 30 = 525$

$$Y - X = 555 \rightarrow (1)$$

Let,

The number of cold tablets available in shop C = $3X - 29 - Y + 24 = 3X - Y - 5$

The number of cold tablets available in shop D = $Y + 787 - 3X + 29 = Y - 3X + 816$

The number of cold tablets available in shop D is 155 more than that of shop C

$$Y - 3X + 816 - (3X - Y - 5) = 155$$

$$2Y - 6X + 821 = 155$$

$$6X - 2Y = 666 \rightarrow (2)$$

$$(1) * 2$$

$$2Y - 2X = 1110$$

Then, sum of equations (1) and (2)

$$6X - 2X = 666 + 1110$$

$$4X = 1776$$

$$X = 444$$

And,

From (1)

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$$Y - 444 = 555$$

$$Y = 999$$

In shop A,

$$\text{The number of cold tablets available} = X + 6 = 444 + 6 = 450$$

$$\text{The number of fever tablets available} = X + 84 = 444 + 84 = 528$$

$$\text{The number of pain tablets available} = [(450 + 528)/(100 - 18.5)] * 18.5 = [978/81.5] * 18.5 = 222$$

$$\text{The total number of tablets available} = 450 + 528 + 222 = 1200$$

In shop B,

$$\text{The number of cold tablets available} = Y - 24 - X - 6 = 999 - 444 - 30 = 525$$

$$\text{The number of fever tablets available} = X + 309 - X - 84 = 225$$

$$\text{The number of pain tablets available} = [(525 + 225)/(100 - 28.57)] * 28.57 = [750/(5/7)] * (2/7) = 300$$

$$\text{The total number of tablets available} = 525 + 225 + 300 = 1050$$

In shop C,

$$\text{The number of cold tablets available} = 3X - 29 - Y + 24 = 1332 - 999 - 5 = 328$$

$$\text{The number of fever tablets available} = Y - X - 309 = 999 - 444 - 309 = 246$$

$$\text{The number of pain tablets available} = [(328 + 246)/(100 - 12.5)] * 12.5 = [574/87.5] * 12.5 = 82$$

$$\text{The total number of tablets available} = 328 + 246 + 82 = 656$$

In shop D,

$$\text{The number of cold tablets available} = Y + 787 - 3X + 29 = 999 - 1332 + 816 = 483$$

$$\text{The number of fever tablets available} = 2Y - 382 - Y = 999 - 382 = 617$$

$$\text{The number of pain tablets available} = [(483 + 617)/(100 - 12)] * 12 = [1100/88] * 12 = 150$$

$$\text{The total number of tablets available} = 483 + 617 + 150 = 1250$$

Let table,

Shop	The number of cold tablets available	The number of fever tablets available	The number of pain tablets available	The total number of tablets available
A	450	528	222	1200
B	525	225	300	1050
C	328	246	82	656
D	483	617	150	1250

Answer: C

The number of cold tablets available in shop E = $(450 + 525 + 483)/3 = 1458/3 = 486$

The number of fever tablets available in shop E = 133.33% of $246 = 328$

The number of pain tablets available in shop E = $150/5 * 7 = 210$

The total number of tablets available in shop E = $486 + 328 + 210 = 1024$

Thus, difference = $1024 - 656 = 368$

6. Questions

Let assume the speed of Praveen, Nakul, Vamsi, and Suriya is 'P' m/sec, 'N' m/sec, 'V' m/sec, and 'S' m/sec respectively.

The total distance between points A and B = 5 km = 5000 m

After 9 minutes starting the race, Praveen travelled 600 m more than Nakul.

$$9 * 60 * P - 9 * 60 * N = 600$$

$$540P - 540N = 600$$

$$P - N = 10/9 \rightarrow (1)$$

After 12 minutes starting the race, Praveen travelled 1.2 km more than Vamsi

$$12 * 60 * P - 12 * 60 * V = 1.2 * 1000$$

$$720P - 720V = 1200$$

$$P - V = 5/3 \rightarrow (2)$$

The time taken by Nakul finish the race is 100 sec earlier than that of Vamsi.

$$(5000/V) - (5000/N) = 100$$

$$(50/V) - (50/N) = 1$$

$$50[(1/V) - (1/N)] = 1$$

$$50 * [(N - V)/NV] = 1$$

$$50 * (N - V) = NV \rightarrow (3)$$

Solve (1) and (2)

$$P - N - P + V = (10/9) - (5/3)$$

$$V - N = -5/9$$

$$N - V = 5/9 \rightarrow (4)$$

Equation (4) substitute into (3)

$$50 * 5/9 = NV$$

$$250/9 = NV$$

$$N = 250/9V \rightarrow (5)$$

@QuizByMickey

Equation (5) substitute into (4)

$$(250/9V) - V = 5/9$$

$$250 - 9V^2 = 5V$$

$$9V^2 + 5V - 250 = 0$$

$$9V^2 + 50V - 45V - 250 = 0$$

$$9V(V - 5) + 50(V - 5) = 0$$

$$(9V + 50)(V - 5) = 0$$

$$V = 5 \text{ and } -50/9$$

$$V = 5 \text{ (positive value)}$$

Then,

From (5)

$$N = 250/45$$

$$N = 50/9$$

And,

From (2)

$$P - V = 5/3$$

$$P - 5 = 5/3$$

$$P = 5 + 5/3$$

$$P = 20/3$$

So,

The speed of Praveen = $(20/3)$ m/sec

The speed of Nakul = $(50/9)$ m/sec

The speed of Vamsi = 5 m/sec

Then,

The time taken by Vamsi to travel 4.5 km = $4500/5 = 900$ seconds

The time taken by Suriya to travel 4.5 km = $(900/5) * 3 = 540$ seconds

The speed of suriya = $4500/540 = (25/3)$ m/sec

Let,

@QuizByMickey

Person	Speed (m/sec)	Speed (km/hr)
Praveen	20/3	$20/3 * (18/5) = 24$
Nakul	50/9	$50/9 * (18/5) = 20$
Vamsi	5	$5 * (18/5) = 18$
Suriya	25/3	$25/3 * (18/5) = 30$

Answer: B

According to the question,

The time taken by Suriya reaches the point B = $5000/(25/3) = 600$ seconds

At this time

The distance travelled by Vamsi = $5 * 600 = 3000$ m

The distance travelled by Praveen = $(20/3) * 600 = 4000$ m

Thus, difference = $4000 - 3000 = 1000$ m = 1 km

7. Questions

Let assume the speed of Praveen, Nakul, Vamsi, and Suriya is 'P' m/sec, 'N' m/sec, 'V' m/sec, and 'S' m/sec respectively.

The total distance between points A and B = 5 km = 5000 m

After 9 of minutes starting the race, Praveen travelled 600 m more than Nakul.

$$9 * 60 * P - 9 * 60 * N = 600$$

$$540P - 540N = 600$$

$$P - N = 10/9 \rightarrow (1)$$

After 12 minutes starting the race, Praveen travelled 1.2 km more than Vamsi

$$12 * 60 * P - 12 * 60 * V = 1.2 * 1000$$

$$720P - 720V = 1200$$

$$P - V = 5/3 \rightarrow (2)$$

The time taken by Nakul finish the race is 100 sec earlier than that of Vamsi.

$$(5000/V) - (5000/N) = 100$$

$$(50/V) - (50/N) = 1$$

$$50[(1/V) - (1/N)] = 1$$

$$50 * [(N - V)/NV] = 1$$

$$50 * (N - V) = NV \rightarrow (3)$$

Solve (1) and (2)

@QuizByMickey

$$P - N - P + V = (10/9) - (5/3)$$

$$V - N = -5/9$$

$$N - V = 5/9 \rightarrow (4)$$

Equation (4) substitute into (3)

$$50 * 5/9 = NV$$

$$250/9 = NV$$

$$N = 250/9V \rightarrow (5)$$

Equation (5) substitute into (4)

$$(250/9V) - V = 5/9$$

$$250 - 9V^2 = 5V$$

$$9V^2 + 5V - 250 = 0$$

$$9V^2 + 50V - 45V - 250 = 0$$

$$9V(V - 5) + 50(V - 5) = 0$$

$$(9V + 50)(V - 5) = 0$$

$$V = 5 \text{ and } -50/9$$

$$V = 5 \text{ (positive value)}$$

Then,

From (5)

$$N = 250/45$$

$$N = 50/9$$

And,

From (2)

$$P - V = 5/3$$

$$P - 5 = 5/3$$

$$P = 5 + 5/3$$

$$P = 20/3$$

So,

The speed of Praveen = $(20/3)$ m/sec

The speed of Nakul = $(50/9)$ m/sec

The speed of Vamsi = 5 m/sec

Then,

@QuizByMickey

The time taken by Vamsi to travel 4.5 km = $4500/5 = 900$ seconds

The time taken by Suriya to travel 4.5 km = $(900/5) * 3 = 540$ seconds

The speed of suriya = $4500/540 = (25/3)$ m/sec

Let,

Person	Speed (m/sec)	Speed (km/hr)
Praveen	$20/3$	$20/3 * (18/5) = 24$
Nakul	$50/9$	$50/9 * (18/5) = 20$
Vamsi	5	$5 * (18/5) = 18$
Suriya	$25/3$	$25/3 * (18/5) = 30$

Answer: A

The distance travelled by Nakul in 9 minutes = $50/9 * 9 * 60 = 3000$ m

The remaining distance = $5000 - 3000 = 2000$ m

The increasing speed of Nakul cycle = 120% of $50/9 = (20/3)$ m/sec

The time taken by Nakul to reach the finishing point B = $9 * 60 + 2000/(20/3) = 540 + 300 = 840$ seconds

$X = 840/3600$ hours

Let,

The distance travelled by Vamsi in 'X' hour = $18 * 840/3600 = 4.2$ km

8. Questions

Let assume the speed of Praveen, Nakul, Vamsi, and Suriya is 'P' m/sec, 'N' m/sec, 'V' m/sec, and 'S' m/sec respectively.

The total distance between points A and B = 5 km = 5000 m

After 9 of minutes starting the race, Praveen travelled 600 m more than Nakul.

$$9 * 60 * P - 9 * 60 * N = 600$$

$$540P - 540N = 600$$

$$P - N = 10/9 \rightarrow (1)$$

@QuizByMickey

After 12 minutes starting the race, Praveen travelled 1.2 km more than Vamsi

$$12 * 60 * P - 12 * 60 * V = 1.2 * 1000$$

$$720P - 720V = 1200$$

$$P - V = 5/3 \rightarrow (2)$$

The time taken by Nakul finish the race is 100 sec earlier than that of Vamsi.

$$(5000/V) - (5000/N) = 100$$

$$(50/V) - (50/N) = 1$$

$$50[(1/V) - (1/N)] = 1$$

$$50 * [(N - V)/NV] = 1$$

$$50 * (N - V) = NV \rightarrow (3)$$

Solve (1) and (2)

$$P - N - P + V = (10/9) - (5/3)$$

$$V - N = -5/9$$

$$N - V = 5/9 \rightarrow (4)$$

Equation (4) substitute into (3)

$$50 * 5/9 = NV$$

$$250/9 = NV$$

$$N = 250/9V \rightarrow (5)$$

Equation (5) substitute into (4)

$$(250/9V) - V = 5/9$$

$$250 - 9V^2 = 5V$$

$$9V^2 + 5V - 250 = 0$$

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$$9V(V - 5) + 50(V - 5) = 0$$

$$(9V + 50)(V - 5) = 0$$

$$V = 5 \text{ and } -50/9$$

$$V = 5 \text{ (positive value)}$$

Then,

From (5)

$$N = 250/45$$

$$N = 50/9$$

And,

From (2)

$$P - V = 5/3$$

$$P - 5 = 5/3$$

$$P = 5 + 5/3$$

$$P = 20/3$$

@QuizByMickey

So,

The speed of Praveen = $(20/3)$ m/sec

The speed of Nakul = $(50/9)$ m/sec

The speed of Vamsi = 5 m/sec

Then,

The time taken by Vamsi to travel 4.5 km = $4500/5 = 900$ seconds

The time taken by Suriya to travel 4.5 km = $(900/5) * 3 = 540$ seconds

The speed of suriya = $4500/540 = (25/3)$ m/sec

Let,

Person	Speed (m/sec)	Speed (km/hr)
Praveen	$20/3$	$20/3 * (18/5) = 24$
Nakul	$50/9$	$50/9 * (18/5) = 20$
Vamsi	5	$5 * (18/5) = 18$
Suriya	$25/3$	$25/3 * (18/5) = 30$

Answer: E

The time taken by Nakul to travel half of the racing distance = $2500/(50/9) = 450$ seconds

The time taken by suriya to travel half of the racing distance = $2500/(25/3) = 300$ seconds

Thus, ratio = $450: 300 = 3: 2$

9. Questions

Let assume the speed of Praveen, Nakul, Vamsi, and Suriya is 'P' m/sec, 'N' m/sec, 'V' m/sec, and 'S' m/sec respectively.

The total distance between points A and B = 5 km = 5000 m

After 9 of minutes starting the race, Praveen travelled 600 m more than Nakul.

$$9 * 60 * P - 9 * 60 * N = 600$$

$$540P - 540N = 600$$

$$P - N = 10/9 \rightarrow (1)$$

After 12 minutes starting the race, Praveen travelled 1.2 km more than Vamsi

$$12 * 60 * P - 12 * 60 * V = 1.2 * 1000$$

$$720P - 720V = 1200$$

$$P - V = 5/3 \rightarrow (2)$$

The time taken by Nakul finish the race is 100 sec earlier than that of Vamsi.

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$$(5000/V) - (5000/N) = 100$$

$$(50/V) - (50/N) = 1$$

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$$50 * [(N - V)/NV] = 1$$

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Solve (1) and (2)

$$P - N - P + V = (10/9) - (5/3)$$

$$V - N = -5/9$$

$$N - V = 5/9 \rightarrow (4)$$

Equation (4) substitute into (3)

$$50 * 5/9 = NV$$

$$250/9 = NV$$

$$N = 250/9V \rightarrow (5)$$

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$$(250/9V) - V = 5/9$$

$$250 - 9V^2 = 5V$$

$$9V^2 + 5V - 250 = 0$$

$$9V^2 + 50V - 45V - 250 = 0$$

$$9V(V - 5) + 50(V - 5) = 0$$

$$(9V + 50)(V - 5) = 0$$

$$V = 5 \text{ and } -50/9$$

$$V = 5 \text{ (positive value)}$$

Then,

From (5)

$$N = 250/45$$

$$N = 50/9$$

And,

From (2)

$$P - V = 5/3$$

$$P - 5 = 5/3$$

$$P = 5 + 5/3$$

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$$P = 20/3$$

So,

$$\text{The speed of Praveen} = (20/3) \text{ m/sec}$$

$$\text{The speed of Nakul} = (50/9) \text{ m/sec}$$

$$\text{The speed of Vamsi} = 5 \text{ m/sec}$$

Then,

$$\text{The time taken by Vamsi to travel 4.5 km} = 4500/5 = 900 \text{ seconds}$$

$$\text{The time taken by Suriya to travel 4.5 km} = (900/5) * 3 = 540 \text{ seconds}$$

$$\text{The speed of suriya} = 4500/540 = (25/3) \text{ m/sec}$$

Let,

Person	Speed (m/sec)	Speed (km/hr)
Praveen	$20/3$	$20/3 * (18/5) = 24$
Nakul	$50/9$	$50/9 * (18/5) = 20$
Vamsi	5	$5 * (18/5) = 18$
Suriya	$25/3$	$25/3 * (18/5) = 30$

Answer: D

$$\text{The speed of Adhi's cycle} = [20/3 + 50/9]/2 = 110/18 \text{ m/sec}$$

$$\text{The distance travelled by Adhi in 360 seconds} = 110/18 * 360 = 2200 \text{ m}$$

$$N = (2200/5000) * 100$$

$$N = 44$$

Then,

$$\text{The required value} = N + 3.5N = 44 + 3.5 * 44 = 44 + 154 = 198.$$

10. Questions

Let assume the speed of Praveen, Nakul, Vamsi, and Suriya is 'P' m/sec, 'N' m/sec, 'V' m/sec, and 'S' m/sec respectively.

$$\text{The total distance between points A and B} = 5 \text{ km} = 5000 \text{ m}$$

After 9 of minutes starting the race, Praveen travelled 600 m more than Nakul.

$$9 * 60 * P - 9 * 60 * N = 600$$

$$540P - 540N = 600$$

$$P - N = 10/9 \rightarrow (1)$$

After 12 minutes starting the race, Praveen travelled 1.2 km more than Vamsi

$$12 * 60 * P - 12 * 60 * V = 1.2 * 1000$$

$$720P - 720V = 1200$$

$$P - V = 5/3 \rightarrow (2)$$

The time taken by Nakul finish the race is 100 sec earlier than that of Vamsi.

$$(5000/V) - (5000/N) = 100$$

$$(50/V) - (50/N) = 1$$

$$50[(1/V) - (1/N)] = 1$$

$$50 * [(N - V)/NV] = 1$$

$$50 * (N - V) = NV \rightarrow (3)$$

Solve (1) and (2)

$$P - N - P + V = (10/9) - (5/3)$$

$$V - N = -5/9$$

$$N - V = 5/9 \rightarrow (4)$$

Equation (4) substitute into (3)

$$50 * 5/9 = NV$$

$$250/9 = NV$$

$$N = 250/9V \rightarrow (5)$$

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$$(250/9V) - V = 5/9$$

$$250 - 9V^2 = 5V$$

$$9V^2 + 5V - 250 = 0$$

$$9V^2 + 50V - 45V - 250 = 0$$

$$9V(V - 5) + 50(V - 5) = 0$$

$$(9V + 50)(V - 5) = 0$$

$$V = 5 \text{ and } -50/9$$

$$V = 5 \text{ (positive value)}$$

Then,

From (5)

$$N = 250/45$$

$$N = 50/9$$

And,

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From (2)

$$P - V = 5/3$$

$$P - 5 = 5/3$$

$$P = 5 + 5/3$$

$$P = 20/3$$

So,

The speed of Praveen = $(20/3)$ m/sec

The speed of Nakul = $(50/9)$ m/sec

The speed of Vamsi = 5 m/sec

Then,

The time taken by Vamsi to travel 4.5 km = $4500/5 = 900$ seconds

The time taken by Suriya to travel 4.5 km = $(900/5) * 3 = 540$ seconds

The speed of suriya = $4500/540 = (25/3)$ m/sec

Let,

Person	Speed (m/sec)	Speed (km/hr)
Praveen	$20/3$	$20/3 * (18/5) = 24$
Nakul	$50/9$	$50/9 * (18/5) = 20$
Vamsi	5	$5 * (18/5) = 18$
Suriya	$25/3$	$25/3 * (18/5) = 30$

Answer: E

The distance travelled by Lara in first 3 min = $20 * 5/18 * 3 * 60 = 1000$ m

The distance travelled by Lara in after 4 min = $120\% \text{ of } 50/9 * 4 * 60 = 1600$ m

The remaining distance = $5000 - 1000 - 1600 = 2400$ m

The time taken by Lara in remaining distance = $2400/(150\% \text{ of } 20/3) = 240$ seconds = 4 minutes

The total time taken by Lara to finish the race = $3 + 4 + 4 = 11$ min

The total time taken by Praveen to finish the race = $5000/(20/3) = 750$ seconds = 12.5 min

The total time taken by Nakul to finish the race = $5000/(50/9) = 900$ seconds = 15 min

The total time taken by Vamsi to finish the race = $5000/5 = 1000$ seconds = $16 (2/3)$ min

The total time taken by Suriya to finish the race = $5000/(25/3) = 600$ seconds = 10 min

So, Suriya is first person to finish the race.

11. Questions

Per unit cost is the first composite number. So, its Rs. 4.

Refrigerator,

The running time of a product (hr) in a day = 60% of 24 = 14.4 hours

The total electricity consumption (units) in July = 22.5

The electricity consumption (units) per day = $22.5/14.4 = 1.5625$

The refrigerator electricity bill for July in Nani's home = $22.5 * 4 = \text{Rs. } 90$

TV,

The running time of a product (hr) in a day = 40% of 24 = 9.6 hours

The total electricity consumption (units) in July = 30

The electricity consumption (units) per day = $30/9.6 = 3.125$

The refrigerator electricity bill for July in Nani's home = $30 * 4 = \text{Rs. } 120$

Mobile charger,

The running time of a product (hr) in a day = 25% of 24 = 6 hours

The total electricity consumption (units) in July = 7.5

The electricity consumption (units) per day = $7.5/6 = 1.25$

The refrigerator electricity bill for July in Nani's home = $7.5 * 4 = \text{Rs. } 30$

Lights,

The running time of a product (hr) in a day = 20% of 24 = 4.8 hours

The total electricity consumption (units) in July = 60

The electricity consumption (units) per day = $60/4.8 = 12.5$

The refrigerator electricity bill for July in Nani's home = $60 * 4 = \text{Rs. } 240$

Modem,

The running time of a product (hr) in a day = 75% of 24 = 18 hours

The total electricity consumption (units) in July = 15

The electricity consumption (units) per day = $15/18 = 5/6$

The refrigerator electricity bill for July in Nani's home = $15 * 4 = \text{Rs. } 60$

The total electricity bill for July in Nani's home = $60 + 240 + 30 + 120 + 90 = \text{Rs. } 540$

Let table,

@QuizByMickey

Product	The running time of a product (hr) in a day	The total electricity consumption (units) in July	The electricity consumption (units) per day	The product electricity bill of July in Nani's home (Rs.)
Refrigerator	14.4	22.5	1.5625	90
TV	9.6	30	3.125	120
Mobile charger	6	7.5	1.25	30
Lights	4.8	60	12.5	240
Modem	18	15	5/6	60

Answer: E

According to the question,

The total electricity bill of June in Nani's home = $1215 - 540 = \text{Rs. } 675$

The total electricity consumption (units) in June = $675/4 = 168.75$

$x\% = [(168.75 - 135)/168.75] * 100$

$x\% = (33.75/168.75) * 100$

$x\% = 20\%$

$x = 20$

12. Questions

Per unit cost is the first composite number. So, its Rs. 4.

Refrigerator,

The running time of a product (hr) in a day = 60% of $24 = 14.4$ hours

The total electricity consumption (units) in July = 22.5

The electricity consumption (units) per day = $22.5/14.4 = 1.5625$

The refrigerator electricity bill for July in Nani's home = $22.5 * 4 = \text{Rs. } 90$

TV,

The running time of a product (hr) in a day = 40% of $24 = 9.6$ hours

The total electricity consumption (units) in July = 30

The electricity consumption (units) per day = $30/9.6 = 3.125$

The refrigerator electricity bill for July in Nani's home = $30 * 4 = \text{Rs. } 120$

Mobile charger,

The running time of a product (hr) in a day = 25% of $24 = 6$ hours

The total electricity consumption (units) in July = 7.5

The electricity consumption (units) per day = $7.5/6 = 1.25$

The refrigerator electricity bill for July in Nani's home = $7.5 * 4 = \text{Rs. } 30$

Lights,

The running time of a product (hr) in a day = 20% of 24 = 4.8 hours

The total electricity consumption (units) in July = 60

The electricity consumption (units) per day = $60/4.8 = 12.5$

The refrigerator electricity bill for July in Nani's home = $60 * 4 = \text{Rs. } 240$

Modem,

The running time of a product (hr) in a day = 75% of 24 = 18 hours

The total electricity consumption (units) in July = 15

The electricity consumption (units) per day = $15/18 = 5/6$

The refrigerator electricity bill for July in Nani's home = $15 * 4 = \text{Rs. } 60$

The total electricity bill for July in Nani's home = $60 + 240 + 30 + 120 + 90 = \text{Rs. } 540$

Let table,

Product	The running time of a product (hr) in a day	The total electricity consumption (units) in July	The electricity consumption (units) per day	The product electricity bill of July in Nani's home (Rs.)
Refrigerator	14.4	22.5	1.5625	90
TV	9.6	30	3.125	120
Mobile charger	6	7.5	1.25	30
Lights	4.8	60	12.5	240
Modem	18	15	5/6	60

Answer: A

The running time of AC (hr) in a day = 75% of 9.6 = 7.2 hours

The electricity consumption per day of AC = 5 units

The total electricity consumption of AC = $7.2 * 5 = 36$

Thus, difference = $60 - 36 = 24$ units

13. Questions

Per unit cost is the first composite number. So, its Rs. 4.

Refrigerator,

The running time of a product (hr) in a day = 60% of 24 = 14.4 hours

The total electricity consumption (units) in July = 22.5

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The running time of a product (hr) in a day = 25% of 24 = 6 hours

The total electricity consumption (units) in July = 7.5

The electricity consumption (units) per day = $7.5/6 = 1.25$

The refrigerator electricity bill for July in Nani's home = $7.5 * 4 = \text{Rs. } 30$

Lights,

The running time of a product (hr) in a day = 20% of 24 = 4.8 hours

The total electricity consumption (units) in July = 60

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The refrigerator electricity bill for July in Nani's home = $60 * 4 = \text{Rs. } 240$

Modem,

The running time of a product (hr) in a day = 75% of 24 = 18 hours

The total electricity consumption (units) in July = 15

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The total electricity bill for July in Nani's home = $60 + 240 + 30 + 120 + 90 = \text{Rs. } 540$

Let table,

@QuizByMickey

Product	The running time of a product (hr) in a day	The total electricity consumption (units) in July	The electricity consumption (units) per day	The product electricity bill of July in Nani's home (Rs.)
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TV	9.6	30	3.125	120
Mobile charger	6	7.5	1.25	30
Lights	4.8	60	12.5	240
Modem	18	15	5/6	60

Answer: D

Thus, ratio = $(1.5625) : (3.125 - 1.25) = 1.5625 : 1.875 = 5 : 6$

14. Questions

Per unit cost is the first composite number. So, its Rs. 4.

Refrigerator,

The running time of a product (hr) in a day = 60% of 24 = 14.4 hours

The total electricity consumption (units) in July = 22.5

The electricity consumption (units) per day = $22.5/14.4 = 1.5625$

The refrigerator electricity bill for July in Nani's home = $22.5 * 4 = \text{Rs. } 90$

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The running time of a product (hr) in a day = 40% of 24 = 9.6 hours

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The running time of a product (hr) in a day = 25% of 24 = 6 hours

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The refrigerator electricity bill for July in Nani's home = $7.5 * 4 = \text{Rs. } 30$

Lights,

The running time of a product (hr) in a day = 20% of 24 = 4.8 hours

The total electricity consumption (units) in July = 60

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Modem,

The running time of a product (hr) in a day = 75% of $24 = 18$ hours

The total electricity consumption (units) in July = 15

The electricity consumption (units) per day = $15/18 = 5/6$

The refrigerator electricity bill for July in Nani's home = $15 * 4 = \text{Rs. } 60$

The total electricity bill for July in Nani's home = $60 + 240 + 30 + 120 + 90 = \text{Rs. } 540$

Let table,

Product	The running time of a product (hr) in a day	The total electricity consumption (units) in July	The electricity consumption (units) per day	The product electricity bill of July in Nani's home (Rs.)
Refrigerator	14.4	22.5	1.5625	90
TV	9.6	30	3.125	120
Mobile charger	6	7.5	1.25	30
Lights	4.8	60	12.5	240
Modem	18	15	5/6	60

Answer: E

The total electricity consumption (units) of Kavi's home in July = 82% of $135 = 110.7$

The total electricity consumption (units) of Mani's home in July = 133.33% of $110.7 = (4/3) * 110.7 = 147.6$

Thus, average = $(135 + 110.7 + 147.6)/3 = 393.3/3 = 131.1$

15. Questions

Per unit cost is the first composite number. So, its Rs. 4.

Refrigerator,

The running time of a product (hr) in a day = 60% of $24 = 14.4$ hours

The total electricity consumption (units) in July = 22.5

The electricity consumption (units) per day = $22.5/14.4 = 1.5625$

The refrigerator electricity bill for July in Nani's home = $22.5 * 4 = \text{Rs. } 90$

TV,

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The running time of a product (hr) in a day = 40% of 24 = 9.6 hours

The total electricity consumption (units) in July = 30

The electricity consumption (units) per day = $30/9.6 = 3.125$

The refrigerator electricity bill for July in Nani's home = $30 * 4 = \text{Rs. } 120$

Mobile charger,

The running time of a product (hr) in a day = 25% of 24 = 6 hours

The total electricity consumption (units) in July = 7.5

The electricity consumption (units) per day = $7.5/6 = 1.25$

The refrigerator electricity bill for July in Nani's home = $7.5 * 4 = \text{Rs. } 30$

Lights,

The running time of a product (hr) in a day = 20% of 24 = 4.8 hours

The total electricity consumption (units) in July = 60

The electricity consumption (units) per day = $60/4.8 = 12.5$

The refrigerator electricity bill for July in Nani's home = $60 * 4 = \text{Rs. } 240$

Modem,

The running time of a product (hr) in a day = 75% of 24 = 18 hours

The total electricity consumption (units) in July = 15

The electricity consumption (units) per day = $15/18 = 5/6$

The refrigerator electricity bill for July in Nani's home = $15 * 4 = \text{Rs. } 60$

The total electricity bill for July in Nani's home = $60 + 240 + 30 + 120 + 90 = \text{Rs. } 540$

Let table,

Product	The running time of a product (hr) in a day	The total electricity consumption (units) in July	The electricity consumption (units) per day	The product electricity bill of July in Nani's home (Rs.)
Refrigerator	14.4	22.5	1.5625	90
TV	9.6	30	3.125	120
Mobile charger	6	7.5	1.25	30
Lights	4.8	60	12.5	240
Modem	18	15	5/6	60

Answer: B

A July month has 31 days,

The total running time of all products (hr) in July = $31 * (14.4 + 9.6 + 6 + 4.8 + 18) = 31 * 52.8 = 1636.8$

16. Questions

Answer: C

The number of boys in class A = $4x$

The number of girls in class A = $3x$

The number of boys in class B = $3y$

The number of girls in class B = $2y$

The number of boys in two classes together = $4x + 3y$

The number of girls in two classes together = $3x + 2y$

Let,

The number of boys in competition = $(1/10) * (4x + 3y)$

The number of girls in competition = $(1/5) * (3x + 2y)$

Let,

$$\{[(1/10) * (4x + 3y)]C_1\} / \{[(1/5) * (3x + 2y) + (1/10) * (4x + 3y)]C_1\} = 5/12$$

$$\{[(1/10) * (4x + 3y)]C_1\} / \{[(1/10) * (6x + 4y + 4x + 3y)]C_1\} = 5/12$$

$$(4x + 3y)/(10x + 7y) = 5/12$$

$$48x + 36y = 50x + 35y$$

$$2x = y$$

The total number of students in the two classes together is 85,

$$4x + 3x + 3 * 2x + 2 * 2x = 85$$

$$4x + 3x + 6x + 4x = 85$$

$$17x = 85$$

$$x = 5$$

The number of boys in class A = $4x = 4 * 5 = 20$

17. Questions

Answer: E

The time taken by A, B, and C together to complete the work = $5(145/151)$ days = $900/151$ days

The time taken by A and B together to complete the work = $7(19/23)$ days = $180/23$ days

The total work = LCM of 900 and 180 = 900

The efficiency of (A + B + C) = $900/(900/151) = 151$

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The efficiency of $(A + B) = 900/(180/23) = 115$

The efficiency of $C = 151 - 115 = 36$

The efficiency of $A = 36/2 * 5 = 90$

The efficiency of $B = 115 - 90 = 25$

Let,

$$(36 + 25) * 5 + (90 + 25) * t + 25 * (t + 7) = 900$$

$$305 + 115t + 25t + 175 = 900$$

$$140t = 420$$

$$t = 3$$

Then,

$$t^4 - t^3 = 3^4 - 3^3 = 81 - 27 = 54$$

18. Questions

Answer: A

The total weight of 18 boys $= 18 * 5x = 90x$ kg

The total weight of 22 girls $= 22 * 4x = 88x$ kg

The total weight of the class $= 90x + 88x = 178x$ kg

The average weight of the class $= (178x)/(18 + 22) = 4.45x$ kg

Let,

$$178x - 34x = (18 + 22 - 10) * (4.45x + 1.75)$$

$$178x - 34x = 30 * (4.45x + 1.75)$$

$$144x = 133.5x + 52.5$$

$$105x = 52.5$$

$$x = 5$$

Let,

The number of boys who left the class $= 10/2 = 5$

The number of girls who left the class $= 5$

The total weight of the students who left the class $= 34x = 170$ kg

The total weight of the girls who left the class $= 170 - 5 * (3x + 4) = 170 - 5 * 19 = 75$ kg

19. Questions

Answer: D

The length, breadth, and height of the rectangular prism $= 14:12:15 = 14x:12x:15x$

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The total surface area of the rectangular prism = $2 * (lb + bh + lh)$

$$2 * (14x * 12x + 12x * 15x + 15x * 14x) = 4464$$

$$168x^2 + 180x^2 + 210x^2 = 2232$$

$$558x^2 = 2232$$

$$x^2 = 4$$

$$x = 2$$

The length, breadth, and height of the rectangular prism is 28 m, 24 m and 30 m respectively.

Then,

The radius of the cone = 28 cm

The height of the cone = 24 m

The volume of the cone = $(1/3) * \pi r^2 h$

$$\text{The volume of the cone} = (1/3) * (22/7) * 28^2 * 24 = 19712 \text{ m}^3$$

20. Questions

Answer: A

The total age of father, mother and son in a family = 140 years

The present age of son = 'x' years

The present age of father = $(30 + x)$ years

The present age of mother = $[(30 + x)/29] * 27 = (810 + 27x)/29$ years

Let,

$$x + 30 + x + (810 + 27x)/29 = 140$$

$$58x + 870 + 810 + 27x = 4060$$

$$85x = 2380$$

$$x = 28 \text{ years}$$

The present age of son = 28 years

The present age of the grandfather = $3 * 28 = 84$ years.

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